Scientific Computing AOL

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Nomor 1

Year (y)	Temperature (x, °C)
1981	14.1999
1983	14.2411
1985	14.0342
1987	14.2696
1989	14.197
1991	14.3055
1993	14.1853
1995	14.3577
1997	14.4187
1999	14.3438

a. Estimate the temperature in even years by linear, quadratic, and cubic interpolation order! Choose the method that you think is appropriate, and explain the difference.

1. a. Linear Interpolation → $Y = Y_1 + (X - X_1)\frac{(Y_2 - Y_1)}{(X_2 - X_1)}$

$X = 1982$, Y?
$X_1 = 1981$, $Y_1 = 14{,}1999$
$X_2 = 1983$, $Y_2 = 14{,}2411$

→ selalu 1, dalam semua input $(x - x_1) = 1$

$Y = 14{,}1999 + (1982 - 1981)\frac{(14{,}2411 - 14{,}1999)}{(1983 - 1982)}$ → Selalu 2, dalam semua input $(x_2 - x_1) = 2$

$Y = 14{,}1999 + 1\,\frac{0.0412}{2} = 14.2205$

$x = 1984$, Y?
$X_1 = 1983$, $Y_1 = 14{,}2411$
$X_2 = 1985$, $Y_2 = 14{,}0342$

$Y = 14{,}2411 + (1984 - 1983)\frac{(14{,}0342 - 14{,}2411)}{(1985 - 1983)}$

$Y = 14{,}2411 - \frac{0.2069}{2} = 14.13765$

$x = 1986$, Y?
$X_1 = 1985$, $Y_1 = 14{,}0342$
$X_2 = 1987$, $Y_2 = 14{,}2696$

$\boxed{\frac{Y_2 - Y_1}{2}} = \frac{0.2354}{2} = 0.1177$

$Y = 14{,}0342 + 0.1177 = 14{,}1519$

$X = 1988$, Y?
$X_1 = 1987$, $Y_1 = 14{,}2696$
$X_2 = 1989$, $Y_2 = 14{,}197$

$\frac{Y_2 - Y_1}{2} = -0.0363$

$Y = 14{,}2696 - 0.0363 = 14{,}2333$

$X = 1990$, Y?
$X_1 = 1989$, $Y_1 = 14{,}197$
$X_2 = 1991$, $Y_2 = 14{,}3055$

$\frac{Y_2 - Y_1}{2} = 0.05425$

$Y = 14{,}197 + 0.05425 = 14{,}25125$

$X = 1992$, Y?
$X_1 = 1991$, $Y_1 = 14{,}3055$
$X_2 = 1993$, $Y_2 = 14{,}1853$

$\frac{Y_2 - Y_1}{2} = -0.0601$

$Y = 14{,}3055 - 0.0601 = 14{,}2454$

$x = 1994, Y?$

$x_1 = 1993, Y_1 = 14,1853$
$x_2 = 1995, Y_2 = 14,3577$ $\Big\} \frac{Y_2 - Y_1}{2} = 0.0862$

$Y = 14,1853 + 0.0862 = 14,2715$

$x = 1996, Y?$

$x_1 = 1995, Y_1 = 14,3577$
$x_2 = 1997, Y_2 = 14,4187$ $\Big\} \frac{Y_2 - Y_1}{2} = 0.0305$

$Y = 14.3577 + 0.0305 = 14,3882$

$x = 1998, Y?$

$x_1 = 1997, Y_1 = 14.4187$
$x_2 = 1999, Y_2 = 14.3438$ $\Big\} \frac{Y_2 - Y_1}{2} = -0.03745$

$Y = 14,4187 - 0.03745 = 14,38125$

Quadratic Interpolation $\to$ $$\boxed{Y = \text{linear}(x_2, x_1, x) + \underbrace{\frac{\left(\frac{Y_3 - Y_2}{x_3 - x_2}\right) - \left(\frac{Y_2 - Y_1}{x_2 - x_1}\right)}{(x_3 - x_1)}}_{f(x_3, x_2, x_1)} \cdot \underbrace{(x - x_1)}_{1}\underbrace{(x - x_2)}_{-1}}$$

$\boxed{1 \quad -1} = (-1)$

$x = 1982 \to x_3 = 1985,$
$x_2 = 1983$
$x_1 = 1981$

$Y = \text{linear}(1983, 1981, 1982) + f(1985, 1983, 1981) \cdot \underbrace{(1982 - 1981)}_{\to 1} \cdot \underbrace{(1982 - 1983)}_{\to -1}$

$Y = 14.2205 + (-0.0310125) \cdot (1)(-1) = 14.2515125$

$x = 1984 \to x_3 = 1987, x_2 = 1985, x_1 = 1981$

$Y = \text{linear}(1985, 1983, 1984) + f(1987, 1985, 1983) \cdot (1)(-1)$
$Y = 14.13765 + 0.0352875(1)(-1) = 14.0823625$

$x = 1986 \to x_3 = 1989, x_2 = 1987, x_1 = 1985$

$Y = \text{linear}(1987, 1985, 1986) + f(1989, 1987, 1985)(1)(-1)$
$Y = 14.1519 + (-0.0385)(-1) = 14.1904$

$x = 1988 \to x_3 = 1991, x_2 = 1989, x_1 = 1987$

$Y = \text{linear}(1989, 1987, 1988) + f(1991, 1989, 1987)(1)(-1)$
$Y = 14.2333 + 0.0226375(-1) = 14.2106625$

$x = 1990 \to x_3 = 1993, x_2 = 1991, x_1 = 1989$

$Y = \text{linear}(1991, 1989, 1990) + f(1993, 1991, 1989)(1)(-1)$
$= 14.25125 + (-0.0285875)(-1) = 14.2798375$

$x = 1992 \rightarrow x_3 = 1995, x_2 = 1993, x_1 = 1991$

$Y = linear(1993, 1991, 1992) + f(1995, 1993, 1991)(1)(-1)$
$= 14.2454 + 0.036575(-1) = 14.208825$

$x = 1994 \rightarrow x_3 = 1997, x_2 = 1995, x_1 = 1993$

$Y = linear(1995, 1993, 1994) + f(1997, 1995, 1993)(1)(-1)$
$= 14.2715 + (0.019925)(-1) = 14.205425$

$x = 1996 \rightarrow x_3 = 1999, x_2 = 1997, x_1 = 1995$

$Y = linar(1997, 1995, 1996) + f(1999, 1997, 1995)(1)(-1)$
$Y = 14.3882 + (-0.0169875)(1)(-1) = 14.4051875$

Cubic Interpolation $\rightarrow$ $Y = quadrat(x_3, x_2, x_1, x) + f(x_4, x_3, x_2, x_1)\underbrace{(x-x_1)}_{1}\underbrace{(x-x_1)}_{-1}\underbrace{(x-x_2)}_{-3}$

$1982 \rightarrow quadrat(1985, 1983, 1981, 1982) + f(1987, 1985, 1983, 1981)\underbrace{(1982-1981)}_{1}\underbrace{(1982-1983)}_{-1}\underbrace{(1982-1985)}_{-3}$
$14.2515125 + 0.01438(3) = 14.2947$

$1984 \rightarrow quadrat(1987, 1985, 1983, 1984) + f(1989, 1987, 1985, 1983)(1)(-1)(-3)$
$14.6823625 + (-0.01563)(3) = 14.6355$

$1986 \rightarrow quadrat(1989, 1987, 1985, 1986) + f(1991, 1989, 1987, 1985)(1)(-1)(-3)$
$14.1904 + 0.01010(3) = 14.221$

$1988 \rightarrow quadrat(1991, 1989, 1987, 1988) + f(1993, 1991, 1989, 1987)(1)(-1)(-3)$
$14.2106025 + (-0.00854)(3) = 14.185$

$1990 \rightarrow quadrat(1993, 1991, 1989, 1990) + f(1995, 1993, 1991, 1989)(1)(-1)(-3)$
$14.2798375 + 0.01086(3) = 14.3124$

$1992 \rightarrow quadrat(1995, 1993, 1991, 1992) + f(1997, 1995, 1993, 1991)(1)(-1)(-3)$
$14.208825 + (-0.00842)(3) = 14.1836$

$1994 \rightarrow quadrat(1997, 1995, 1993, 1994) + f(1999, 1997, 1995, 1993)(1)(-1)(-3)$
$14.205425 + (-0.00051)(3) = 14.2039$

Menurut saya metode yang paling tepat adalah cubic Interpolation dikarenakan data yang dimiliki itu tidak memiliki jarak yang sama dan cubic interpolation lebih cocok daripada linear interpolation untuk data yang tidak memiliki jarak yang sama. Perbedaan dari ke 3 metode Interpolatio adalah Linear Interpolation lebih cocok untuk digunakan pada data yang memiliki jarak yang sama satu sama lain. Quadratic Interpolation sama seperti Cubic Interpolation itu baik untuk digunakan pada data yang memiliki jarak yang berbeda satu sama lain tetapi salah satu kelemahan quadratic interpolation adalah jika salah satu titik data diubah maka itu akan menyebabakan perubahan yang bayak terhadap kurvanya. Cubic Interpolation lebih baik dari Quadratic karena jika ada perubahan pada salah satu titik maka perubahan kurva akan lebih kecil dan akan berada sekitar titik tersebut.

b. Perform a least-square regression of the above data to estimate the temperature in even year

1.b.

$$m = \frac{[(10 \cdot 283684{,}08) - (19900 \cdot 142{,}5528)]}{[(10 \cdot 39601330) - (19900)^2]} = 0.012155758$$

$$b = \frac{[142{,}5528 - (0.012155758 \cdot 19900)]}{10} = -9.93468$$

$$LSQ = mx + b$$
$$= 0.012155758x + (-9.93468)$$

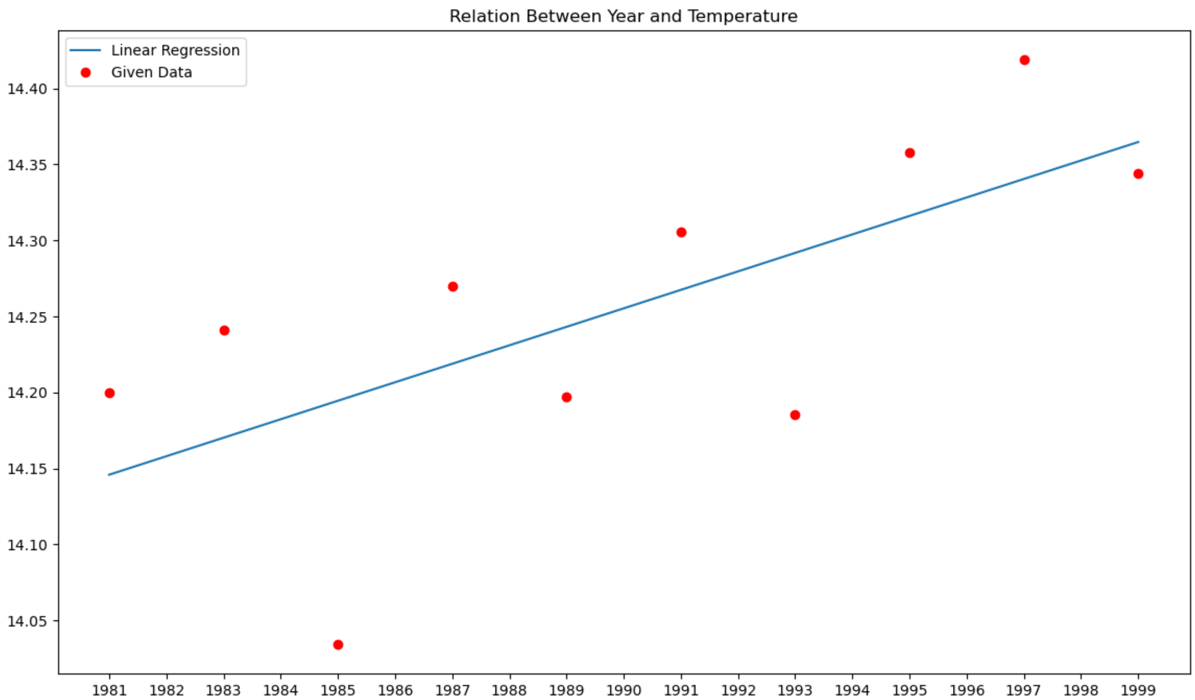
$1982 \rightarrow 0.012155758(1982) - 9.93468 = 14.158$
$1984 \rightarrow 0.012155758(1984) - 9.93468 = 14.1823$
$1986 \rightarrow 0.012155758(1986) - 9.93468 = 14.2067$
$1988 \rightarrow 0.012155758(1988) - 9.93468 = 14.231$
$1990 \rightarrow 0.012155758(1990) - 9.93468 = 14.2553$
$1992 \rightarrow 0.012155758(1992) - 9.93468 = 14.2796$
$1994 \rightarrow 0.012155758(1994) - 9.93468 = 14.3039$
$1996 \rightarrow 0.012155758(1996) - 9.93468 = 14.3282$
$1998 \rightarrow 0.012155758(1998) - 9.93468 = 14.3525$
$2000 \rightarrow 0.012155758(2000) - 9.93468 = 14.3768$

c. Perform an analysis of the difference between the results of the regression and interpolations you can above, explain based on the theoretical basis you have learned.

Perbedaan antara hasil dari regression dan interpolation dapat dijelaskan oleh sifat dan tujan dari ke-2 metode tersebut. Kedua metode ini didesain untuk memprediksi suatu titik diantara 2 titik yang mempunyai suatu jarak. Interpolation lebih didesain untuk memprediksi pada data yang dimana titik-titiknya mempunyai jarak yang sama satu sama lain. Sedangkan Regression lebih didesain untuk memprediksi pada data yang dimana titik-titiknya mempunyai jarak yang tidak sama satu sama lain. Regression juga bertujuan untuk mendapatkan relasi antar independent dan dependent variable, sedangkan Interpolation bertujuan untuk mendapatkan value dependent variable berdasarkan value dari independent variable.

d. Make a plot that describes the relationship between Temperature (y) and Year (x) as informatively as possible for the reader, based on the results of your analysis using Python library.

```python
df = pd.read_excel("./Number-1-data.xlsx")
df.rename(columns={'Year (y)' : 'x', 'Temperature (x, oC)' : 'y'},
inplace=True)

df['x2'] = df['x'] ** 2
df['y2'] = df['y'] ** 2
df['xy'] = df['x'] * df['y']
m = round((10 * sum(df['xy']) - sum(df['y']) * sum(df['x'])) / (10 *
sum(df['x2']) - sum(df['x'])**2), 9)

b = round((sum(df['y']) - m * sum(df['x'])) / 10, 5)

def least_square(m, b, x) :
    return round(m * x + b, 4)

for i in df['x'] :
    print(f"{i + 1} -> {m} x {i + 1} + ({b}) = {least_square(m, b, i + 1)}")
x = np.array(df["x"])
y = np.array(df["y"])

#plotting
plt.figure(figsize = (14, 8))
plt.plot(x, m*x+b, label="Linear Regression")
plt.plot(x, y, "ro",label="Given Data")
plt.title("Relation Between Year and Temperature")
plt.xticks(np.arange(min(x), max(x) + 1, 1))
plt.legend()
plt.show()
```



Nomor 2

a. Write down your manual calculation AND Python script to answer above's question

Manual

2a. Manual Calculation -> taylor expansion Formula

$$\boxed{\hookrightarrow f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} + \frac{f''''(0)x^4}{4!} + \dots + \frac{f^n(0)x^n}{n!}}$$

- $\sin(x)$

$f(x) = \sin(x) \rightarrow f(0) = \sin(0) = 0$
$f'(x) = \cos(x) \rightarrow f'(0) = \cos(0) = 1$
$f''(x) = -\sin(x) \rightarrow f''(0) = -\sin(0) = 0$
$f'''(x) = -\cos(x) \rightarrow f'''(0) = -\cos(0) = -1$
$f''''(x) = \sin(x) \rightarrow f''''(0) = \sin(0) = 0$

Answer: $\sin(x) = x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040}$

- $\cos(x)$

$f(x) = \cos(x) \rightarrow f(0) = \cos(0) = 1$
$f'(x) = -\sin(x) \rightarrow f'(0) = -\sin(0) = 0$
$f''(x) = -\cos(x) \rightarrow f''(0) = -\cos(0) = -1$
$f'''(x) = \sin(x) \rightarrow f'''(0) = \sin(0) = 0$
$f''''(x) = \cos(x) \rightarrow f''''(0) = \cos(0) = 1$

Answer: $\cos(x) = 1 + \left(-\frac{x^2}{2}\right) + \frac{x^4}{24} - \frac{x^6}{720} + \frac{x^8}{40320}$

- $\sin(x)\cos(x)$

$f(x) = \sin(x)\cos(x) \rightarrow f(0) = \sin(0)\cos(0) = 0$
$f'(x) = \cos(2x) \rightarrow f'(0) = \cos(2 \cdot 0) = 1$
$f''(x) = -2\sin x \cos x \rightarrow f''(0) = -2\sin(0)\cos(0) = 0$
$f'''(x) = -2(\cos^2(x) - \sin^2(x)) \rightarrow f'''(0) = -2(\cos^2(0) - \sin^2(0)) = -2$
$f''''(x) = 4\sin(x)\cos(x) \rightarrow f''''(0) = 4\sin x \cos x = 0$

- $\sin(x)\cos(x)$

$f(x) = \sin x \cos x \rightarrow f(0) = \sin(0)\cos(0) = 0$
$f'(x) = \cos(2x) \rightarrow f'(0) = \cos(2 \cdot 0) = 1$
$f''(x) = -2\sin(2x) \rightarrow f''(0) = -2\sin(2 \cdot 0) = 0$
$f'''(x) = -4\cos(2x) \rightarrow f'''(0) = -4\cos(2 \cdot 0) = -4$
$f''''(x) = 8\sin(2x) \rightarrow f''''(0) = 8\sin(2 \cdot 0) = 0$

Answer: $\sin(x)\cos(x) = x - \frac{4x^3}{3!} + \frac{16x^5}{5!} - \frac{64x^7}{7!}$

Script

```
import sympy as sm
x = sm.symbols('x')

def taylor(f, order, x0) :
    t = 0
    for i in range(0, order) :
        t = t + (sm.diff(f, x, i).subs(x, x0) / sm.factorial(i)) * (x ** i)
    return t
sin_formula = taylor(sm.sin(x), 4, 0)
sin_formula
cos_formula = taylor(sm.cos(x), 4, 0)
cos_formula
sincos_formula = taylor(sm.sin(x) * sm.cos(x), 4, 0)
sincos_formula
```

Sin

$$-\frac{x^7}{5040} + \frac{x^5}{120} - \frac{x^3}{6} + x$$

Cos

$$-\frac{x^6}{720} + \frac{x^4}{24} - \frac{x^2}{2} + 1$$

Sin Cos

$$-\frac{4x^7}{315} + \frac{2x^5}{15} - \frac{2x^3}{3} + x$$

b. Which produces less error for x=π/2: computing the Taylor expansion for sin and cos separately then multiplying the result together, or computing the Taylor expansion for the product first then plugging in x?

Script

```python
import sympy as sm

x = sm.symbols('x')
x_value = (np.pi) / 2

# Taylor Expansion Around 0
a = 0

approx_1 = sin_formula.subs(x, x_value) * cos_formula.subs(x, x_value)
approx_2 = sincos_formula.subs(x, x_value)
exact_value = np.sin(x_value) * np.cos(x_value)

error_approx_1 = np.abs(approx_1 - exact_value)
error_approx_2 = np.abs(approx_2 - exact_value)

print("Comparing approach")
print(f"Error of Approach 1: {error_approx_1}")
print(f"Error of Approach 2: {error_approx_2}")
```

Result

```
Comparing approach
Error of Approach 1: 0.000894382649068296
Error of Approach 2: 0.0376103079518116
```

From the result above we can conclude that computing the Taylor expansion for sin and cos separately then multiplying the result together produce less error for x=π/2

c. Use the same order of Taylor series to approximate cos (π/4) and determine the truncation error bound. You may include either your manual calculation OR Python script for this question

```python
x_value = (np.pi) / 4
cos_approx = cos_formula.subs(x, x_value)
exact_value = np.cos(x_value)
error = np.abs(cos_approx - exact_value)

print("Taylor series for cos(X)")
print(f"Exact Value: {exact_value}")
print(f"Approximation: {cos_approx}")
print(f"Error: {error}")
print()
```

```
Taylor series for cos(X)
Exact Value: 0.7071067811865476
Approximation: 0.707103214822846
Error: 0.00000356636370191232
```

Nomor 3

3a. 20 digits $\int_0^{2\pi} x^3 - 0.3x^2 - 8.56x + 8448$

- Riemann Integral
  $\Delta x = (2\pi - 0)/20$
  $\Delta x = \pi/10$

$i = 0$
$x = 0 + 0 \cdot (\pi/10) = 0$
$f(x) = f(0) = 0^3 - 0.3(0)^2 - 8.56(0) + 8448 = 8448$

* $i = 1$
  $x = 0 + 1(\pi/10) = \pi/10$
  $f(x) = f(1) = (\pi/10)^3 - 0.3(\pi/10)^2 - 8.56(\pi/10) + 8448$
* $i = 2$
  $x = 0 + 2(\pi/10) = \pi/5$
  $f(x) = f(2) = (\pi/5)^3 - 0.3(\pi/5)^2 - 8.56(\pi/5) + 8448$
* $i = 3$
  $x = 0 + 3(\pi/10) = 3\pi/10$
  $f(x) = f(3) = (3\pi/10)^3 - 0.3(3\pi/10)^2 - 8.56(3\pi/10) + 8448$
* $i = 4 \to f(4) = (4\pi/10)^3 - 0.3(4\pi/10)^2 - 8.56(4\pi/10) + 8448$
* $i = 5 \to f(5) = (5\pi/10)^3 - 0.3(5\pi/10)^2 - 8.56(5\pi/10) + 8448$
* $i = 6 \to f(6) = (6\pi/10)^3 - 0.3(6\pi/10)^2 - 8.56(6\pi/10 + 8448$
* $i = 7 \to f(7) = (7\pi/10)^3 - 0.3(7\pi/10)^2 - 8.56(7\pi/10 + 8448$
* $i = 8 \to f(8) = (8\pi/10)^3 - 0.3(8\pi/10)^2 - 8.56(8\pi/10 + 8448$
* $i = 9 \to f(9) = (9\pi/10)^3 - 0.3(9\pi/10)^2 - 8.56(9\pi/10 + 8448$
* $i = 10 \to f(10) = (\pi)^3 - 0.3(\pi)^2 - 8.56(\pi) + 8448$
* $i = 11 \to f(11) = (11\pi/10)^3 - 0.3(11\pi/10)^2 - 8.56(11\pi/10) + 8448$
* $i = 12 \to f(12) = (12\pi/10)^3 - 0.3(12\pi/10)^2 - 8.56(12\pi/10) + 8448$
* $i = 13 \to f(13) = (13\pi/10)^3 - 0.3(13\pi/10)^2 - 8.56(13\pi/10) + 8448$
* $i = 14 \to f(14) = (14\pi/10)^3 - 0.3(14\pi/10)^2 - 8.56(14\pi/10) + 8448$
* $i = 15 \to f(15) = (15\pi/10)^3 - 0.3(15\pi/10)^2 - 8.56(15\pi/10) + 8448$
* $i = 16 \to f(16) = (16\pi/10)^3 - 0.3(16\pi/10)^2 - 8.56(16\pi/10) + 8448$
* $i = 17 \to f(17) = (17\pi/10)^3 - 0.3(17\pi/10)^2 - 8.56(17\pi/10) + 8448$
* $i = 18 \to f(18) = (18\pi/10)^3 - 0.3(18\pi/10)^2 - 8.56(18\pi/10) + 8448$
* $i = 19 \to f(19) = (19\pi/10)^3 - 0.3(19\pi/10)^2 - 8.56(19\pi/10) + 8448$
* $i = 20 \to f(20) = (2\pi)^3 - 0.3(2\pi)^2 - 8.56(2\pi) + 8448$

maka $f(x) = f(0) + f(1) + \dots + f(20) = 248{,}47252$

- trapezoid Rule
  $\Delta x = \frac{2\pi - 0}{20} = \frac{\pi}{10}$

$t_n = \frac{\Delta x}{2}[f(x_0) + 2f(x_2) + \dots 2f(x_{n-1}) + f(x_n)]$

$x_0 = 0$
step $= \Delta x$ $\to$ $x_1 = x_0 + \Delta x$
$x_2 = x_1 + \Delta x$
$\vdots$
$x_{20} = x_{19} + \Delta x$

menggunakan $f(x_0) - f(x_{20})$ pada riemann integral maka
$t_n = 229.2454789 67$

- Simpson Rule

$\Delta x = \pi/10$

$$S_n = \frac{\Delta x}{3}\left[f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 4f(x_{19}) + 2f(x_{20})\right]$$

$x_0 = 0$

Stepsize $= \Delta x$

$x_1 = x_0 + \Delta x$

$x_2 = x_1 + \Delta x$

$\vdots$

$x_{20} = x_{19} + \Delta x$

maka $S_n = 240{,}9440649$

3C. Interpolation Polinomial

x	f(x)
-1,1	15.18
-0.3	10.962
0.8	1.920
1,9	-2.040

$Y = L_0 f(x_0) + L_1 f(x_1) + L_2 f(x_2) + L_3 f(x_3)$

$$f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)}Y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)}Y_1 +$$

$$\frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)}Y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}Y_3$$

$$= \frac{(x+0.3)(x-0.8)(x-1.9)}{(-0.8)(-1.9)(-3)}(15.18) + \frac{(x+1.1)(x-0.8)(x-1.9)}{(0.8)(-1.1)(-2.2)}(10.962)$$

$$+ \frac{(x+1.1)(x+0.3)(x-1.9)}{(1.9)(1.1)(-1.1)}(1.92) + \frac{(x+1.1)(x+0.3)(x-0.8)}{(3)(2.2)(1.1)}(-2.04)$$

$$= (x^3 - 2.4x^2 + 0.71x + 0.406)(-3.33) + (x^3 - 1.6x^2 - 1.45x + 1.67)(5.66)$$
$$+ (x^3 - 0.5x^2 - 2.33x - 0.627)(-0.84) + (x^3 + 0.6x^2 - 0.79x - 0.264)(-0.28)$$

$f(x) = 1.21x^3 - 0.812x^2 - 8.3929x + 0.69942$

$f'(x) = 3{,}63x^2 - 1.624x - 8.3929$

$f''(x) = 7.26x - 1.624$